

METHODOLOGY REPORT: STUDY BUDDY

Computer Science II

Marquez W., Duroy P., Morga G., Costo C.

Grade 8 - Jasmine

Dachel P. Raagas

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INTRODUCTION

Study Buddy is a productivity-based academic platform designed specifically for students. The system integrates task management, grade tracking, and quiz generation into a single centralized dashboard. Its goal is to reduce the fragmentation of academic systems and centralize academic monitoring.

This methodology report is intended to document the group's journey in developing the initial frameworks and core features necessary for the intended program, which is Study Buddy.

RESEARCH AND PLANNING PHASE

After the presentation stage, the group entered the initial development stage. They analyzed common academic productivity challenges, including scattered systems and inefficient task monitoring. Sample grading systems, assessment weights, and system flow structures were collected to simulate realistic academic environments. A blueprint of the program's modular framework was created before coding.

METHODOLOGY DEVELOPMENT PHASE:

The group followed an Incremental and Modular development approach to reduce the risk of recurring errors and allow them to obtain early user feedback. Each core feature (Task Manager, Grade Calculator, Quiz Generator) was developed and implemented individually by team members before any system integration.

SYSTEM BLUEPRINT

Our system follows Client-Server Architecture. The server component is implemented using Python and Flask, handling all data processing, grade calculations, and quiz generation. The User Interface (UI)

utilizes HTML, CSS, and JavaScript to manage user interaction and content rendering. JSON files function as a lightweight tool for local data storage (Graphical User Interface/ GUI was utilized).

CORE FEATURES IMPLEMENTATION

Once all the core features were developed separately, all of them were implemented into 1 program, with the use of **Visual Studio Code (VSCode)**.

- 1) Smart Task Manager [**MARQUEZ W**]: Enables subject-based task entry with deadlines stored in JSON format.
- 2) Grade Calculator [**MORGA G. AND COSTO Z.**]: Implements weighted average computation using score, maximum score, and weight inputs for both PSHS and DepEd system grading.
- 3) Quiz Generator [**DUROY P**]: Utilizes structured JSON question banks and random sampling for quiz creation.

Each module underwent unit-level testing before integration. Manual test cases validated weighted grade accuracy, data persistence, and random quiz generation reliability. Error-handling mechanisms were implemented for invalid inputs and missing data scenarios. However, there were still errors despite these precautions.

AREAS FOR IMPROVEMENT

Time management, coordination, and documentation updates were identified as development challenges. Future improvements include database migration (e.g., Liquibase, cloud services), user authentication, enhanced UI responsiveness, and automated testing. This will soon be updated once the 4th Quarter semester starts.

CONCLUSION

The Study Buddy program has undergone multiple system changes and is still developing. However, the implementation of core features has already been done, and further changes will be observed during the 4th Quarter semester.

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